Amazon Sales Data Analysis Project

Overview

This project analyzes Amazon sales data of three cities in Myanmar: Yangon, Mandalay, and Naypyitaw for the first quarter of 2019.

Objective

The main objective is to understand sales performance and customer behavior using data analysis and visualization.

In [5]:
```
import pandas as pd

df = pd.read_csv("Amazon.csv")
df.head()
```

Out[5]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	54
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	8
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	16.2155	34
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	23.2880	48
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	30.2085	63

Group data by City

In [6]:
```
# Group data by 'City' and sum the 'Total' sales
city_sales = df.groupby("City")["Total"].sum()
```

```
# Display city-wise total sales
city_sales
```

Out[6]:
```
City
Mandalay     106197.6720
Naypyitaw    110568.7065
Yangon       106200.3705
Name: Total, dtype: float64
```

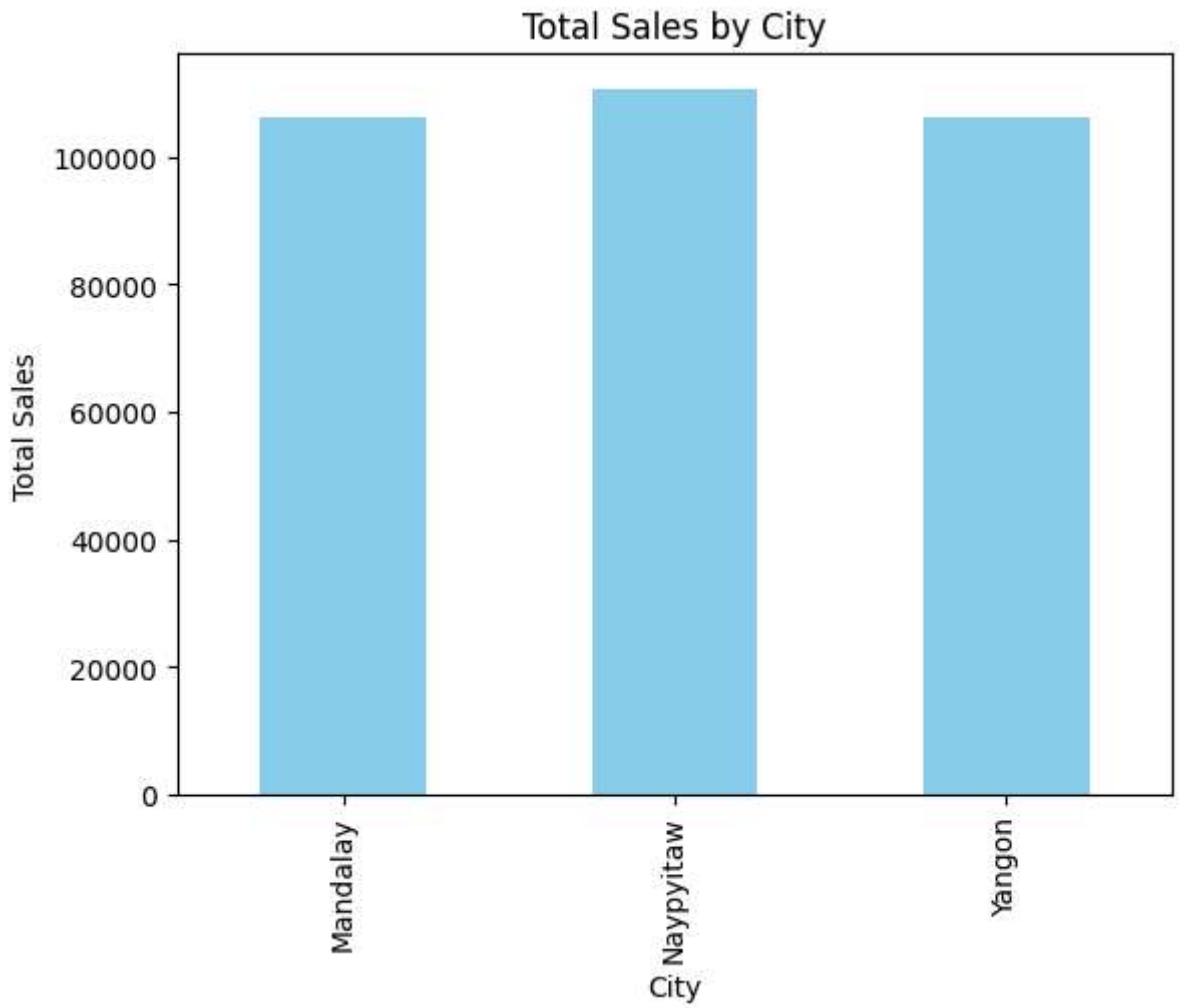
Bar Chart (City-wise Sales)

In [7]:
```
# Import matplotlib for plotting
import matplotlib.pyplot as plt

# Plot bar chart for city-wise total sales
city_sales.plot(kind="bar", color="skyblue")

# Add title and labels
plt.title("Total Sales by City")
plt.xlabel("City")
plt.ylabel("Total Sales")

# Show the plot
plt.show()
```



Payment Method Pie Chart

Objective:

Show which payment method customers use the most.

In [ ]:
```
## Count number of transactions per payment method
```

In [8]:
```
# Count number of transactions for each payment method
payment_count = df["Payment"].value_counts()

# Display the count
payment_count
```

Out[8]:
```
Payment
Ewallet        345
Cash           344
Credit card    311
Name: count, dtype: int64
```

Pie Chart (Payment Method)

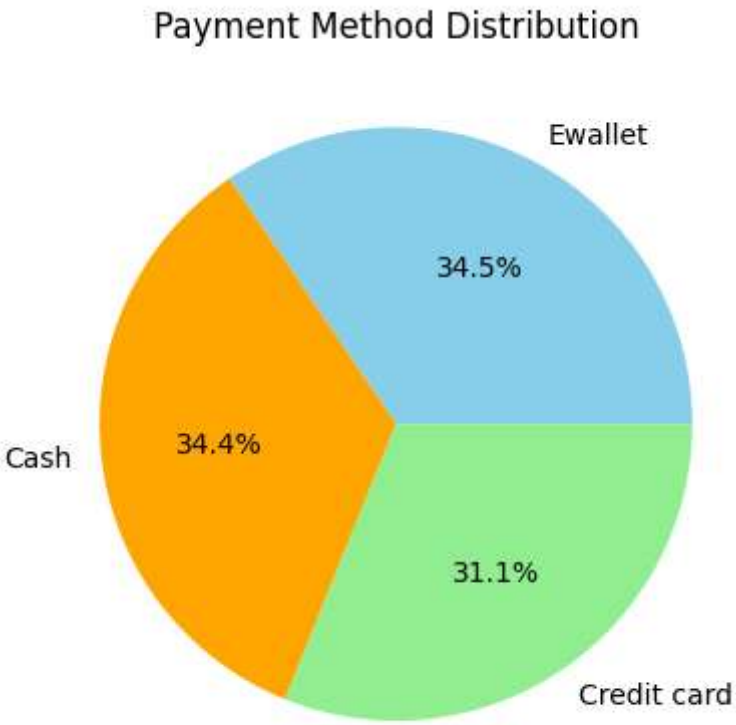
In [9]:
```
# Import matplotlib (agar pehle import nahi kiya)
import matplotlib.pyplot as plt

# Pie chart for payment method distribution
payment_count.plot(kind="pie", autopct="%1.1f%%", colors=["skyblue", "orange", "lig

# Add title
plt.title("Payment Method Distribution")

# Remove y-axis label
plt.ylabel("")

# Show the plot
plt.show()
```



Product Line Revenue Bar Chart

Objective:

Show which product line generates the highest total revenue.

Group data by Product Line

In [10]:
```
# Group data by 'Product line' and sum total revenue
product_sales = df.groupby("Product line")["Total"].sum()

# Display product line revenue
product_sales
```

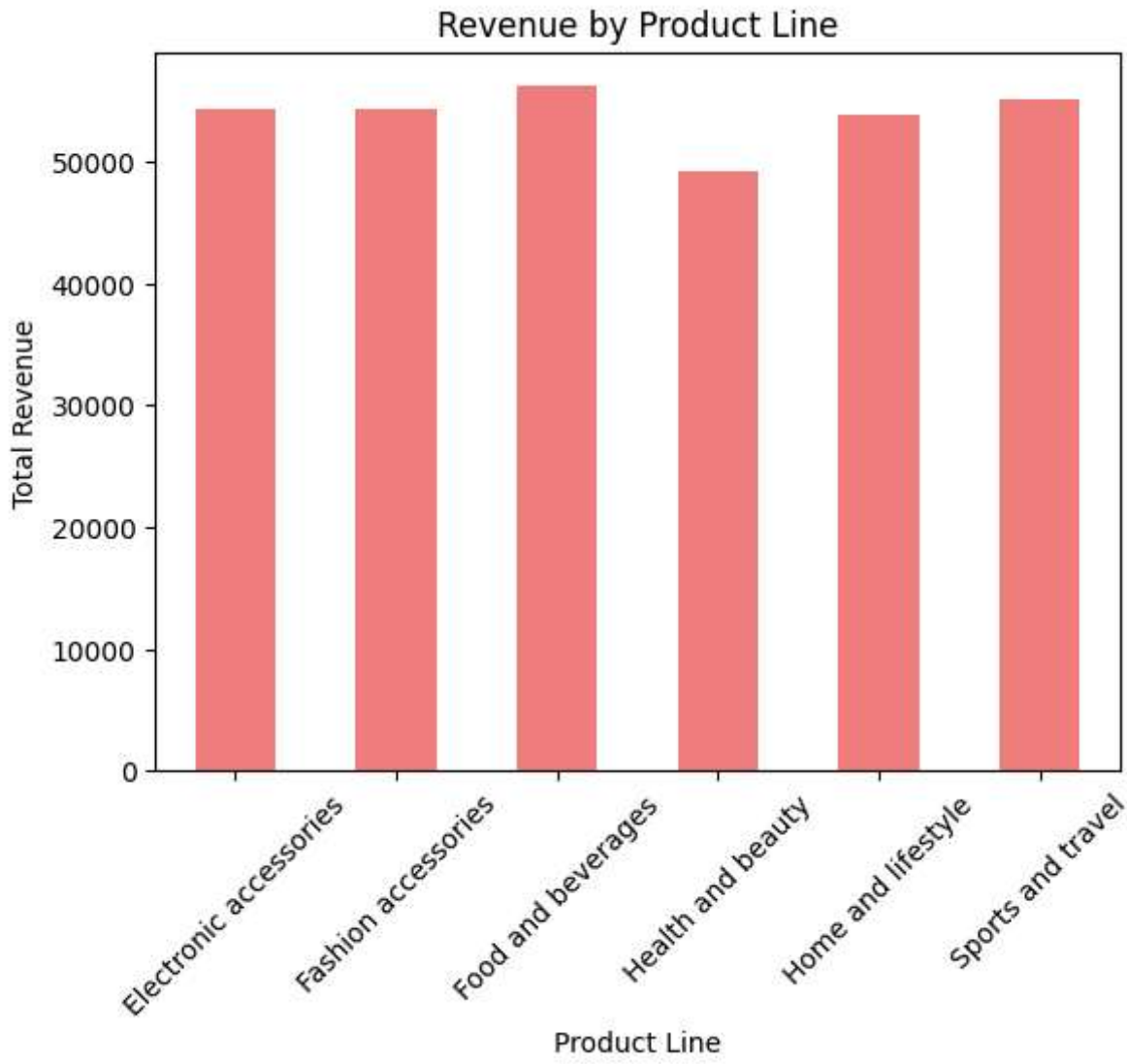
Out[10]:
```
Product line
Electronic accessories    54337.5315
Fashion accessories       54305.8950
Food and beverages        56144.8440
Health and beauty         49193.7390
Home and lifestyle        53861.9130
Sports and travel         55122.8265
Name: Total, dtype: float64
```

Bar Chart for Product Line Revenue

In [12]:
```
# Plot bar chart for product line revenue
product_sales.plot(kind="bar", color="lightcoral")

# Add title and labels
plt.title("Revenue by Product Line")
plt.xlabel("Product Line")
plt.ylabel("Total Revenue")
plt.xticks(rotation=45)  # Rotate x-axis labels for better visibility

# Show the plot
plt.show()
```



Monthly Sales Trend (Line Chart)

Extract Month from Date

In [14]:
```
# Convert 'Date' column to datetime format (day first)
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

# Extract month number
df["Month"] = df["Date"].dt.month

# Check first 5 rows
df[["Date","Month"]].head()
```

Out[14]:

	Date	Month
0	2019-01-05	1
1	2019-03-08	3
2	2019-03-03	3
3	2019-01-27	1
4	2019-02-08	2

Line Chart for Monthly Sales

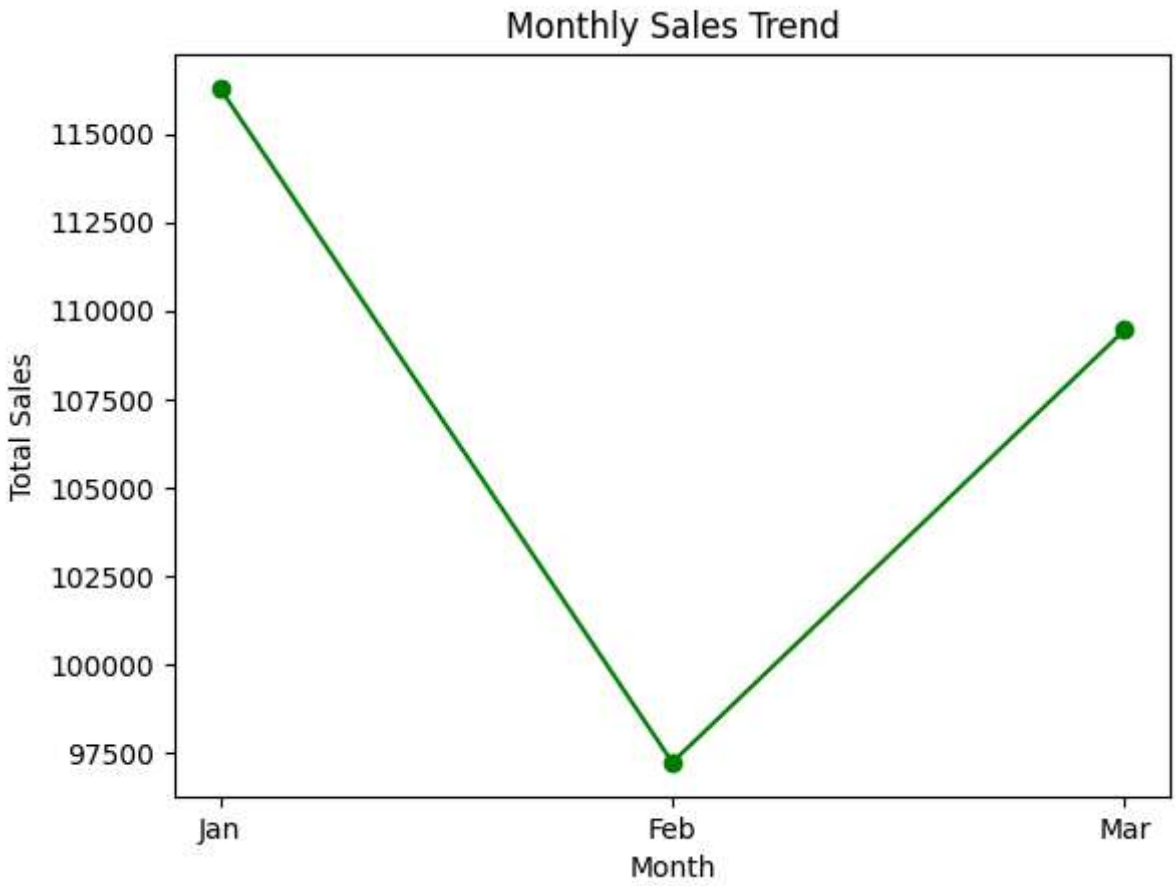
In [15]:

```
# Group total sales by month
monthly_sales = df.groupby("Month")["Total"].sum()

# Plot line chart
monthly_sales.plot(kind="line", marker="o", color="green")

# Add title and labels 3
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks([1,2,3], ["Jan","Feb","Mar"])  # Replace month numbers with names

# Show the plot
plt.show()
```



Insight: Total Sales by City

- Naypyitaw has the highest total sales.
- Yangon and Mandalay have similar sales.

Insight: Total Sales by Payment Method

- Ewallet is the most preferred payment method.
- Cash and Credit Card are used less compared to Ewallet.

Insight: Total Sales by Product Line

- Health and Beauty is the top selling product line.
- Food and Beverages and Sports and Travel also perform well.

Insight: Total Sales by Customer Type

- Member customers generate more revenue than Normal customers.

Conclusion

- Naypyitaw is the highest revenue generating city.
- Health and Beauty is the top selling product line.
- Member customers generate more revenue than Normal customers.
- Ewallet is the most preferred payment method.

This analysis helps understand customer behavior and improve business strategy.